

Eliot Abramo

Embedded Systems Engineer | Space Hardware–Software Co-design | FPGA and Analog Systems

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Embedded systems engineer focused on reliable space hardware–software co-design. Flight-tested electronics for orbital payloads (launch scheduled March 2026). European Space Robotics Champion 2025.

Experience

Embedded Systems Engineer – Space Payloads, SpaceLocker, Toulouse, France Sep. 2025 – Present

Flight hardware–software development for orbital payload systems

- Developed an automated ECSS-compliant reliability-tracing tool integrated into KiCad, providing end-to-end traceability from schematic/BOM to ECSS reliability budgets and informing design trade-offs early in the design cycle.
- Designed, built and tested electronics for space payload systems scheduled for orbital deployment in March 2026, including power conditioning, data handling and watchdog/safing circuitry.
- Participated in official ESA PDR and CDR processes: assembled data packages, justified system design choices and iterated based on formal feedback.
- Designed an analog-logic safety board with voting logic, watchdogs and safe-mode fallbacks to ensure autonomous recovery from satellite anomalies, including dead-on-arrival scenarios.
- Contributed to space qualification testing: TVAC campaigns, vibration and shock testing, functional and integration tests, and end-to-end system validation.
- Wrote flight software in C and Rust under ECSS constraints, using MISRA-inspired coding practices for fault-tolerant communication, on-board diagnostics and telemetry handling.

Lead System Engineer, EPFL Xplore Rover, EPFL Aug. 2024 – Sep. 2025

1st Place – Space Robotics Champions, European Rover Challenge 2025

- Led the electronics and embedded systems side of the rover that won ERC 2025, working across avionics, power, communications, perception and ground segment.
- Coordinated rover system architecture within a large multidisciplinary team, owning electrical and communication interfaces between STM32/ESP32-based avionics, sensors, actuators and the operator station.
- Co-designed and implemented a full custom avionics communication stack: deterministic SPI interconnect with UART failover, zero-copy circular buffers and CRC framing used as the main inter-node bus during ERC operations.
- Built a full-duplex communication bridge linking ESP32-based avionics nodes to a ROS 2 network, using custom-framed packets with explicit CRC error checking and type-safe dispatch between embedded and Linux domains.
- Set up hardware-in-the-loop testbeds and regression test harnesses for avionics and communication stacks to catch integration issues before field tests and competition runs.

Research Assistant, Laboratory of Wave Engineering (LWE), EPFL Sep. 2024 – Aug. 2025

- Led an independent research project on “Nonlinear activation functions for analog neurons in low-power deep learning applications,” exploring physical nonlinearities for analog accelerators under the guidance of a PhD researcher.
- Built and characterized a tunable 2.4 GHz binary metasurface using coupled resonators with PIN-diode control for programmable RF wavefront manipulation.
- Prototyped an optical GELU-like activation using structured incoherent light and Fresnel propagation as a primitive for optical neural networks.
- Co-designed simulations and experiments using CST Studio Suite, vector network analyzers and Python/C++ processing pipelines.

Embedded Systems Engineer, EPFL Xplore Rover, EPFL Sep. 2023 – Aug. 2024

- Designed and manufactured custom PCBs for STM32 and ESP32 avionics with attention to power efficiency, signal integrity and robustness to outdoor competition conditions.
- Implemented and maintained communication stacks over CAN, SPI, I²C and UART, including CRC validation, zero-copy buffers and graceful handling of link failures.
- Developed and used a custom CAN MessageBus framework for static, zero-overhead packet exchange across rover subsystems, with packed packet definitions and protocol versioning.
- Contributed to lower- and higher-level system pipelines in ROS 2, helping integrate avionics data into autonomy and operator tools.

- Managed communication strategy and content, increasing student engagement by around 30%.
- Organized the 2023 IEEEExtreme Hackathon and industry visits, connecting students with engineering teams in industry.

- Assisted on projects involving blockchain integration for IoT systems and microgrid/renewable energy research in collaboration with the University of New Mexico.

Selected Technical Projects

- **Custom ROS 2 ↔ Embedded MCU Bridge** – Full-duplex hardware-agnostic communication bridge between ROS 2 and bare-metal microcontrollers using custom CRC-framed packets. Target implementation over UART/SPI, providing reliable, high-throughput telemetry and command channels between Linux and embedded nodes.
- **Full Custom Avionics Communication Stack** – Deterministic SPI layer with UART failover, zero-copy circular buffers and CRC validation, forming the main real-time inter-node messaging backbone in the EPFL Xplore rover avionics.
- **SpaceEther** – Lightweight custom transport protocol running on raw Ethernet or UDP, offering framing, optional forward error correction and authentication, and topic-oriented payloads for embedded “black boxes” talking directly to standard PCs or ROS 2 systems without extra PC-side software.
- **FPGA + ARM Radio Astronomy Accelerator** – Fully pipelined DSP chain on a Zynq-7020 (PYNQ-Z2) that converts raw radio telescope data into calibrated spectra, extracting Doppler-shifted 21 cm hydrogen lines to estimate Milky Way rotation, with latency/energy trade-offs explored across multiple FFT configurations.
- **CNN FPGA Accelerator** – 3×3 convolution + bias + ReLU fused into a single HLS kernel with multi-row caching and filter parallelism, reaching about 50× speed-up over a software baseline on PYNQ.

Education

École Polytechnique Fédérale de Lausanne (EPFL), Switzerland

- MSc in Electrical and Electronic Engineering – Specialization in Microelectronics 2025 – Present
- BSc in Electrical and Electronic Engineering (Bilingual: English/French) 2022 – 2025
- **Focus areas:** Embedded systems, FPGA/DSP, analog computing, space systems.

Jumeirah English Speaking School (JESS), Dubai, UAE

2018 – 2022

- International Baccalaureate Diploma: **43/45**.
- Extended essay: “How effective are Lagrange polynomials and Fourier series in modeling the spread of COVID-19 in the UK?”

Competitions and Awards

- **European Rover Challenge 2025** – 1st Place, Lead System Engineer for the winning Mars rover team.
- **4x4 in Schools, UAE** – Top 13 nationally; Innovation Award for suspension and obstacle navigation design.
- **First Lego League (FLL) Robotics** – Design and Innovation Award 2020 (250+ teams); Top 10 in 2018 (300+ teams).
- **UN Geneva Forum Presenter** – Presented an ecological sensing device project, highlighting the use of embedded electronics in environmental research.

Technical Skills

- **Space Systems:** ECSS standards, TVAC testing, vibration/shock testing, PDR/CDR workflows, redundancy/safing design, basic EMC/EMI considerations.
- **Programming:** C, C++, Rust, Python (OpenCV, ROS 2, PyTorch), VHDL, MATLAB, Bash, Julia, Java, JavaScript.
- **Embedded & FPGA:** STM32, ESP32, Zynq/PYNQ, Zephyr and FreeRTOS, CAN, SPI, I²C, UART, HIL/SIL testing.
- **Tools:** Vivado/Vitis HLS, KiCad, CST Studio Suite, Git, STM32CubeIDE, Linux, Fusion 360.
- **Languages:** English (fluent), French (fluent), Spanish (conversational).